

GEOLOGIC REVIEW IN DEPOSITIONAL DIPPING YAMPA BASIN (G.R.I.D.D.Y.)

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ABSTRACT

Steamboat Springs, Colorado, lies within a geologically diverse setting characterized by hydrothermal activity, subsurface faulting, glacial deposits, and active fluvial erosion; however, the area remains largely understudied geophysically. Here, we investigate how subsurface faulting and sedimentary deposition interact to control the geological evolution of the Yampa Basin. We present a comprehensive geophysical characterization of the Yampa Basin south of Steamboat Springs using multiphysics methods, including the interpretation of active-source reflection seismic and gravity data. We acquired 13 km of vibroseis seismic data along Routt County Roads 14 and 22 and integrated these data with existing seismic and gravity profiles along Colorado Highway 131. Gravity data were collected using a CG-5 gravimeter and reduced to Bouguer anomalies to isolate subsurface density variations. For detailed seismic processing workflows, including velocity analysis and depth conversions, see the companion abstract by Shallenberger et al.

The Precambrian crystalline basement, composed of dense, erosion-resistant metamorphic rocks, forms the structural foundation of the basin and is expected to produce a strong seismic reflector at depth, allowing us to constrain basin thickness and geometry. Seismic and gravity interpretations to reveal lateral variations in the subsurface structure throughout the Yampa Basin. We expect reflection seismic profiles to resolve distinct stratigraphic horizons and fault geometries at depth, while Bouguer gravity anomalies will constrain bulk density variations associated with basin-bounding fault systems. Together, these datasets will provide the first integrated geophysical characterization of subsurface structure south of Steamboat Springs, with implications for understanding the structural controls on the region's hydrothermal systems and seismic hazard potential.